OHDSI Study Agent for Human–AI Team-Based Real-World Evidence Generation in the VHA

Richard D. Boyce, PhD | University of Pittsburgh • VA VINCI

Study Agent offers reusable, governed AI support for OHDSI research workflows

Service-based ACP/MCP coordination + deterministic validation; researcher confirmation remains central.

WHY / GOVERNANCE

RWE study design involves linked judgments about phenotypes, causal assumptions, cohorts, and interpretation.

RESEARCHER CONTROL

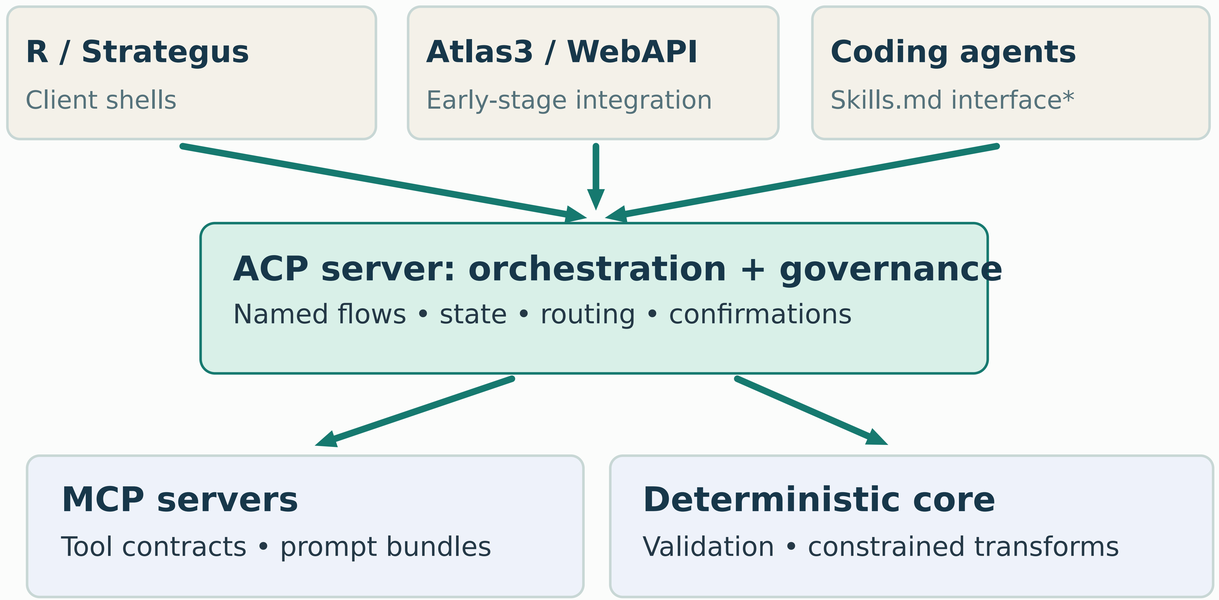
Confirmations and stage gates constrain consequential steps. Model inputs use sanitized summaries and metadata, not row-level clinical data.

PROTOTYPE MATURITY

~TRL 4–5 (report estimate)
Scripted demonstration workflows
No comparative effectiveness or decision-accuracy results

1 SERVICE-BASED ARCHITECTURE

Multiple clients access shared services



* Coding-agent integration: additional design context; verify before publication.

2 SUPPORTED FLOWS

phenotype_recommendation

phenotype_make_computable

concept_sets_review

cohort_critique_general_design

workflow_context_dialogue

OHDSI deterministic code can be reused through bounded flows; specific integrations to verify.

3 WORKFLOW CONTEXT + EXAMPLE

“How should I think about time at risk?”
/ohdsi uses incidence-study stage, role, and selected artifacts to return contextual guidance.

4 DEMONSTRATIONS + EVALUATION

R Strategus prototype; early Atlas3/WebAPI support.
Screenshot slot: authentic demo asset required.
Needs larger validation of accuracy, utility and reliance.

EXHIBITS / HANDOFF

Next: verified architecture diagram; compact named-flow table; authentic application screenshot.
Unverified: coding-agent support and precise OHDSI code reuse. No outcome claims.